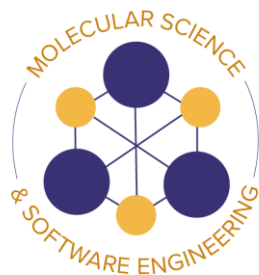


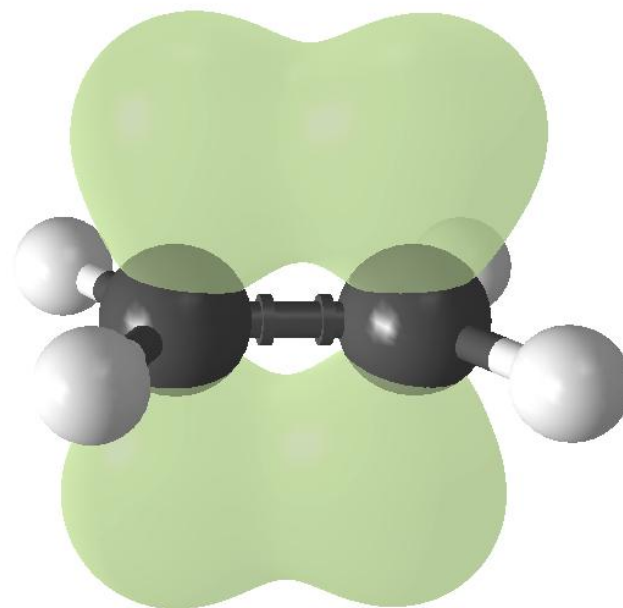
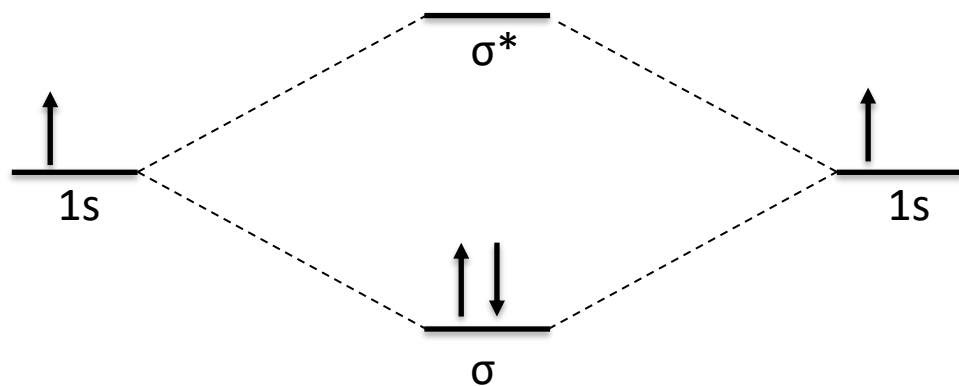
# Electronic Structure Theory

**Berkeley**  
UNIVERSITY OF CALIFORNIA



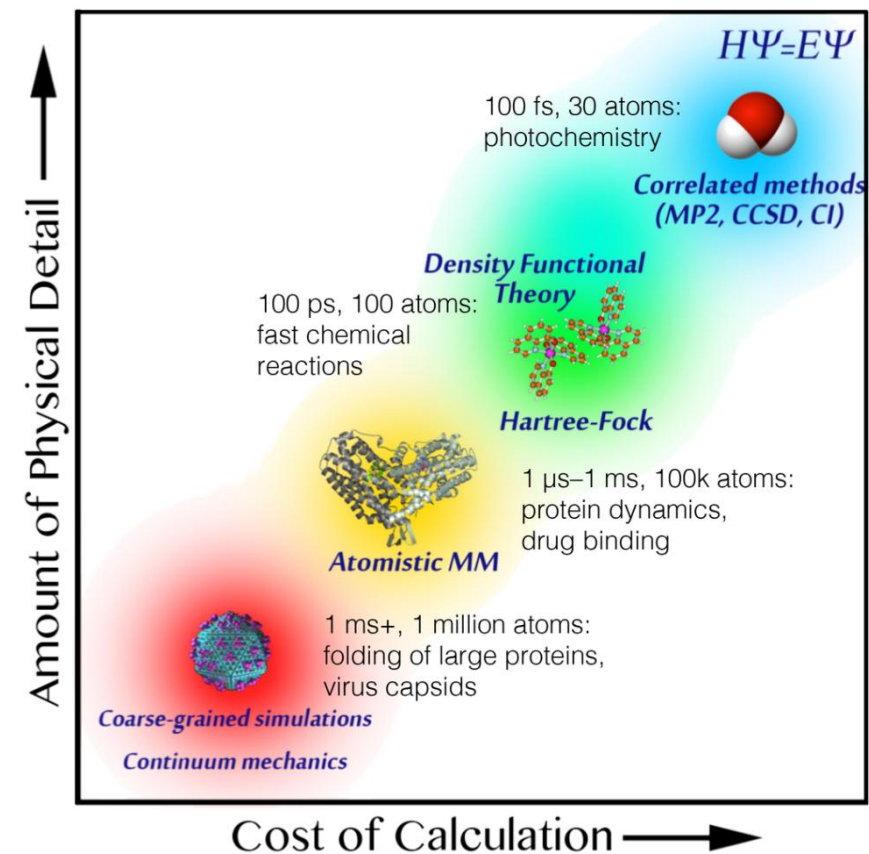
# What is Electronic Structure Theory

- Determination of location and structure of electrons within a molecule

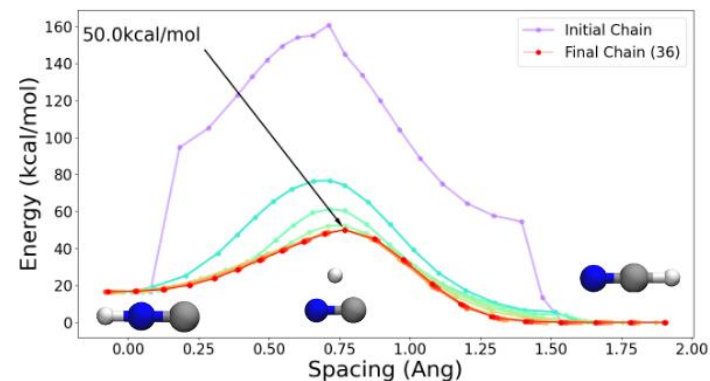
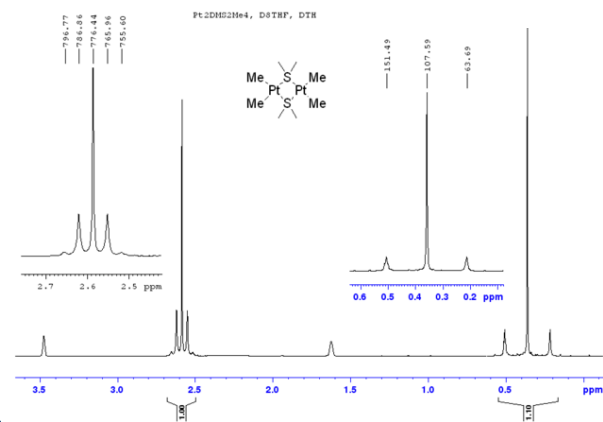
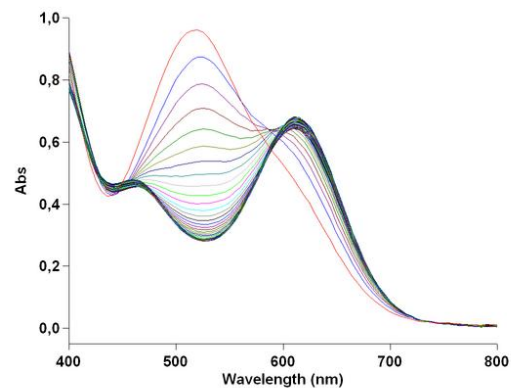
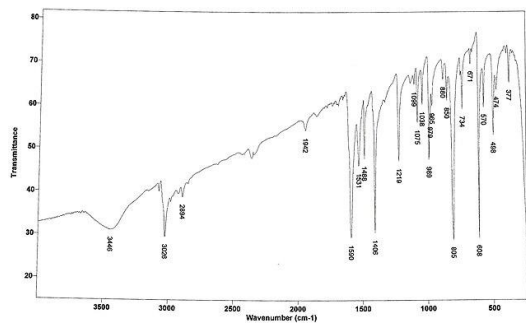


# Quantum vs. Classical Calculations

- The model you choose depends on what you want to calculate
- “All models are wrong, some are useful”

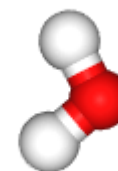


# Singlepoint Calculations



- Calculation on a single fixed geometry
- Electronic properties:
  - UV/Vis, IR, CD spectra
  - NMR/Magnetic susceptibility
  - Bonding information
  - Reaction energies/Transition states

# Geometry Optimization



|   |        |         |        |
|---|--------|---------|--------|
| O | 0.0000 | 0.0000  | 0.0000 |
| H | 0.0000 | 1.5875  | 0.0000 |
| H | 0.7938 | -1.8521 | 0.0000 |

|   |         |         |        |
|---|---------|---------|--------|
| O | -0.1282 | -0.1662 | 0.0000 |
| H | 0.3142  | 0.6933  | 0.0000 |
| H | 0.7087  | -0.7918 | 0.0000 |

# QM Model Selection

- Molecular Orbitals (MOs) are expanded in the basis of Atomic Orbitals (AOs). These AOs are defined by your *basis set*
- Your *method* defines how you are calculating those MOs (what approximations)
- Combined, these are referred to as your *model*, *model chemistry*, or *level of theory*
- Often written together with a slash, with the method first

Method (Hartree-Fock)

hf/sto-3g

b3lyp/6-31g\*

pbe0/def2-SVP

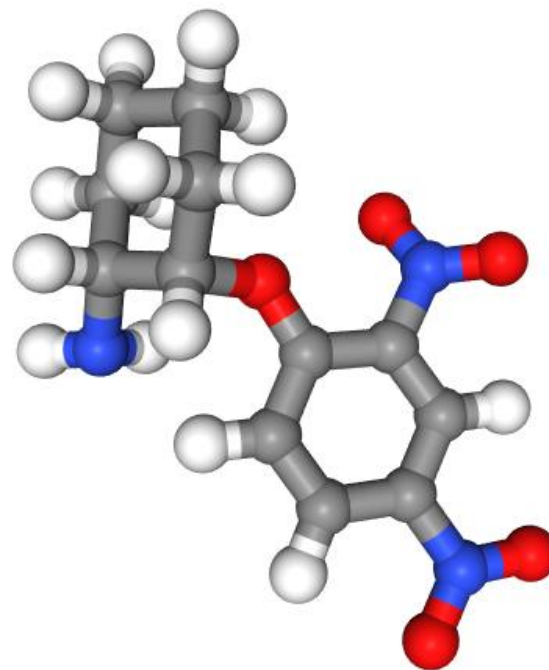
ccsd(t)/aug-cc-pvdz

Basis set

(Slater type orbital fit with 3 gaussians)

# Why not use QM for everything?

- Geometry optimization of this molecule took **7 hours** on two cores (psi4,  $\omega$ B97x/6-31G\*)



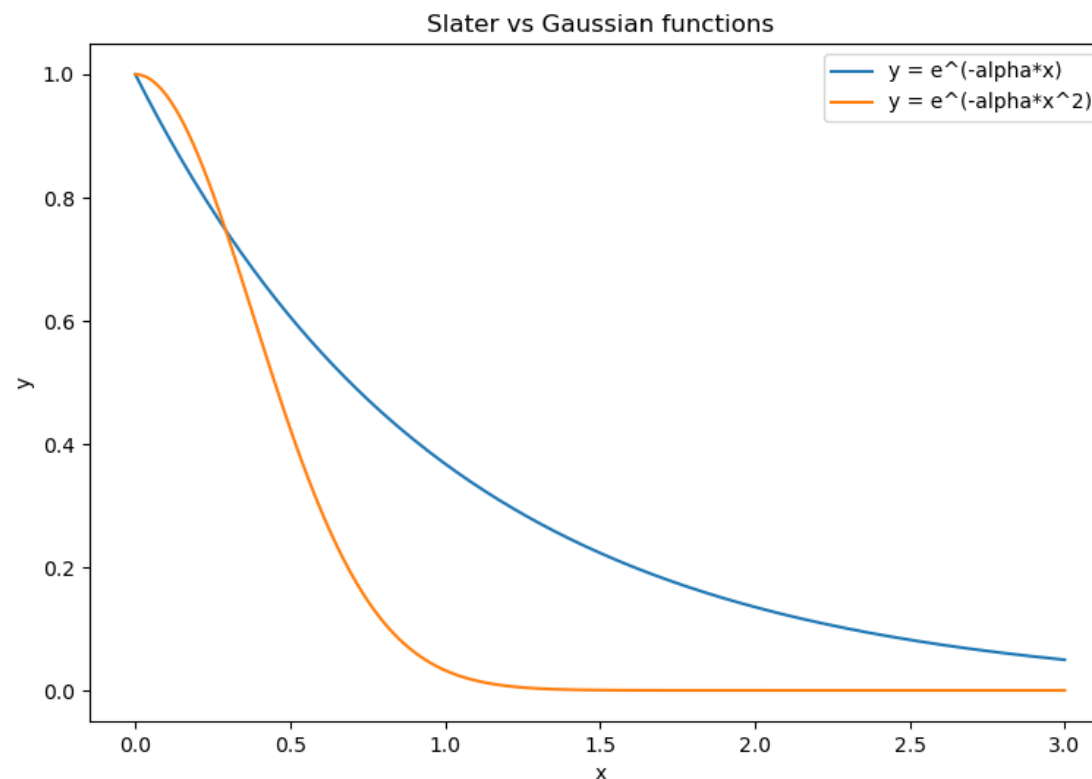
# Method Hierarchy

- There is a zoo of methods, all with their strengths and weaknesses
- Hartree-Fock (HF) – very basic, but inaccurate relative to others (lacks correlation)
- Møller–Plesset perturbation theory (MP2, MPn) – Adds correlation
- Coupled Cluster (CCS, CCSD, CCSD(T)) – Very accurate, very expensive
- Density Functional Theory (DFT) – Very good balance between accuracy and cost
  - Walter Kohn – 1998 Nobel Prize in Chemistry for the development of DFT



# Basis Sets

- Two common types of functions in basis sets
- Slater:  $e^{-\alpha x}$
- Gaussian:  $e^{-\alpha x^2}$
- Most atom centered codes use gaussian
- *In general*, larger basis sets are more accurate (but more expensive of course)



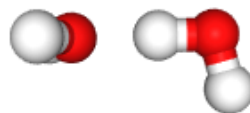
# Molecules

- In QM, we generally only care about:
  - Element (number of protons)
  - Coordinates
  - Charge & multiplicity (unpaired electrons)
- **Bonds do not exist in the QM world**
- Two units of distance
  - Angstrom ( $\text{\AA}$ ,  $10^{-10}\text{m}$ )
  - bohr ( $5.291\,772 \times 10^{-11}\text{m}$  or  $0.5291772\, \text{\AA}$ )

|   |                |                  |                 |
|---|----------------|------------------|-----------------|
| O | 1.548014347149 | 0.00000000000000 | 0.060071441686  |
| H | 0.548241218221 | 0.00000000000000 | 0.038771425885  |
| H | 1.819295099755 | 0.00000000000000 | -0.902428819751 |

# Interaction Energy

- The main output of a singlepoint calculation is the total energy of the molecule.
- This is defined as being relative to all nuclei and all electrons being infinitely far apart. Generally, this energy is not meaningful, but *relative* energies can be useful.



- The *interaction energy* for a dimer (two molecules) is defined as  $IE = E_{AB} - (E_A + E_B)$  where  $E_{AB}$  is the energy of both molecules computed together, and  $E_A, E_B$  are the energies of each of the molecules computed by themselves.